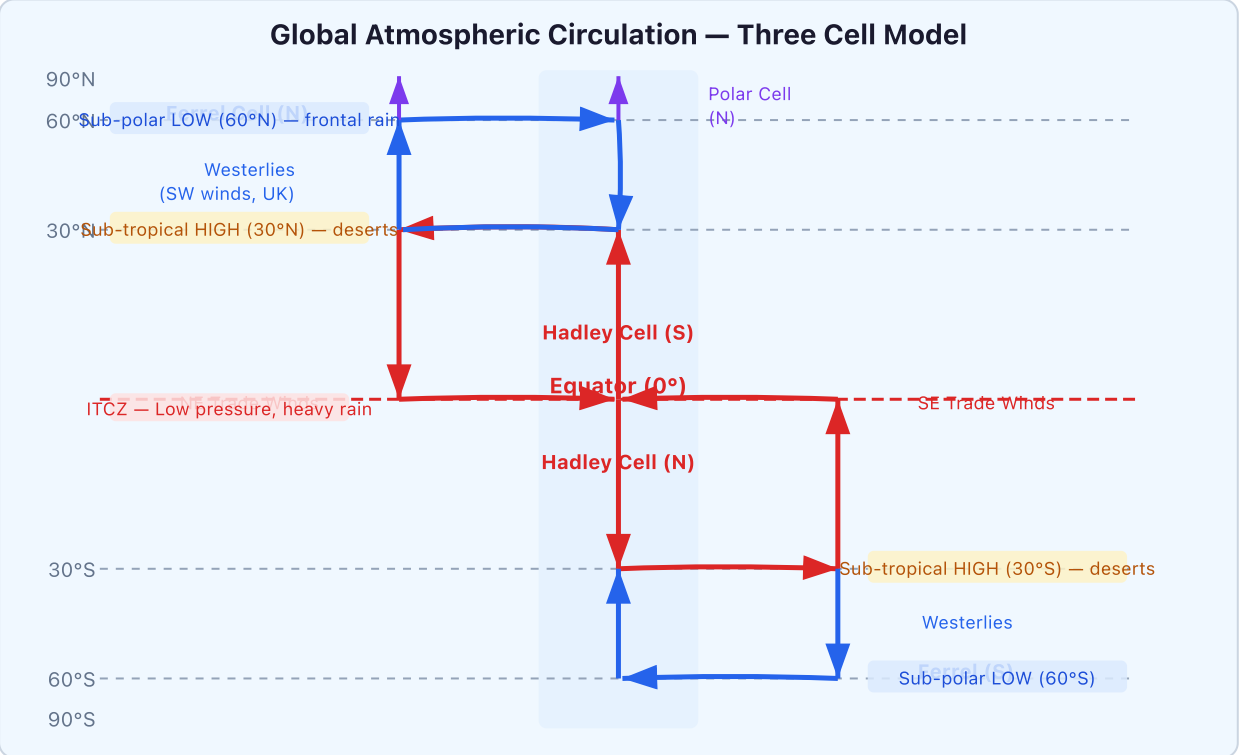
AQA GCSE Geography (8035) — Paper 1: Living with the Physical Environment · 3.1.1 The Challenge of Natural Hazards

1.2 Weather Hazards

1. Global Atmospheric Circulation

Global atmospheric circulation: The large-scale movement of air around the Earth driven by unequal heating. The equator receives more solar radiation than the poles, creating temperature and pressure differences. This drives a system of convection cells (Hadley, Ferrel, and Polar cells) that redistribute heat energy around the globe and create the major wind belts and climate zones.



Pressure Belt / Zone	Latitude	Conditions	Climate result

ITCZ (Inter-Tropical Convergence Zone)	0° (equator)	Rising air, intense heating, low pressure	Equatorial climate: hot, very wet all year; tropical rainforest
Sub-tropical high pressure	~30°N/S	Descending air, high pressure, dry	Hot desert climates (Sahara, Arabian, Australian deserts)
Sub-polar low pressure	~60°N/S	Cold polar air meets warm tropical air; fronts; rising air	Temperate climate with frontal rainfall (UK, NW Europe)
Polar high pressure	90°N/S	Descending cold air, high pressure, very dry	Polar/tundra climate: very cold, low precipitation (mainly snow)

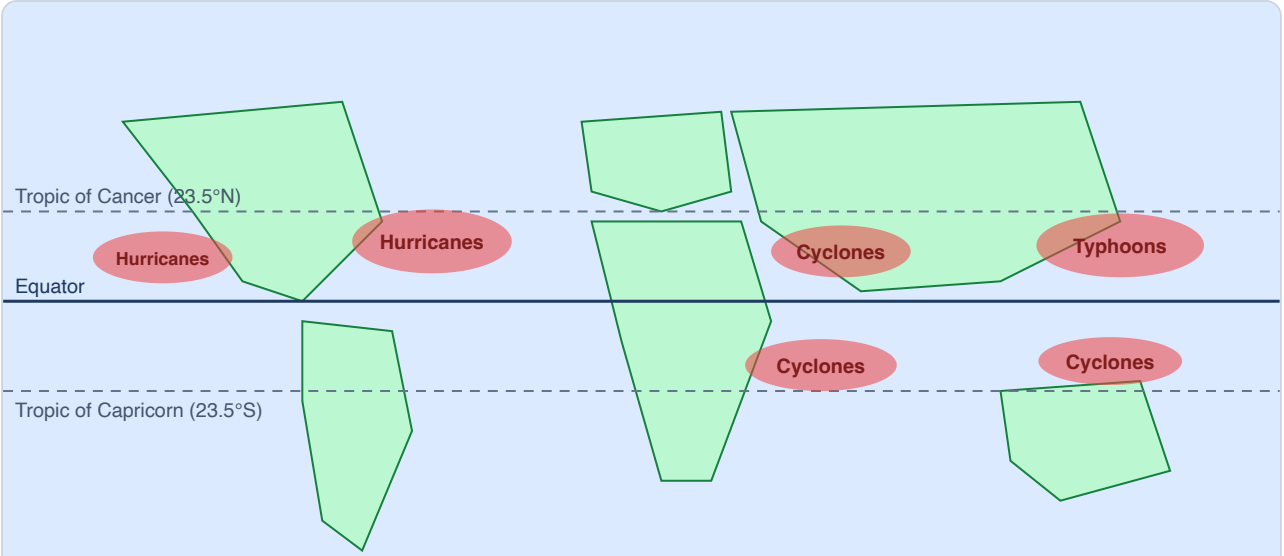
Surface winds

- **Trade winds** blow from the sub-tropical high (30°) back towards the low pressure at the Equator. The Coriolis effect bends them, so they blow from the north-east in the Northern Hemisphere and south-east in the Southern Hemisphere.
- **Westerlies** (south-westerlies over the UK) blow from the sub-tropical high towards the sub-polar low at 60°, bringing mild, wet Atlantic air and depressions to the UK.
- Where trade winds meet at the ITCZ, warm moist air rises, giving heavy convectional rain — the same rising air that helps tropical storms form over warm oceans.

2. Global Distribution of Tropical Storms

Tropical storm — an intense low-pressure weather system with sustained wind speeds above 119 km/h (to count as a hurricane, typhoon or cyclone), heavy rainfall and a calm central eye. They are called **hurricanes** in the North Atlantic and NE Pacific, **typhoons** in the NW Pacific, and **cyclones** in the Indian Ocean and South Pacific.

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Tropical storms form between about 5° and 30° north and south of the Equator (simplified map, not to scale)

- Tropical storms form over **warm tropical oceans**, mostly between **5° and 30° north and south** of the Equator.
- They do not form **within about 5° of the Equator** because the Coriolis effect is too weak there to make the air spin.
- They move **westwards** with the trade winds, then often curve polewards. They **weaken over land** (no supply of warm, moist air) and over cooler water.
- Most form in **late summer and autumn** (June–November in the Northern Hemisphere), when sea temperatures are highest.

3. Causes and Formation of Tropical Storms

Conditions needed

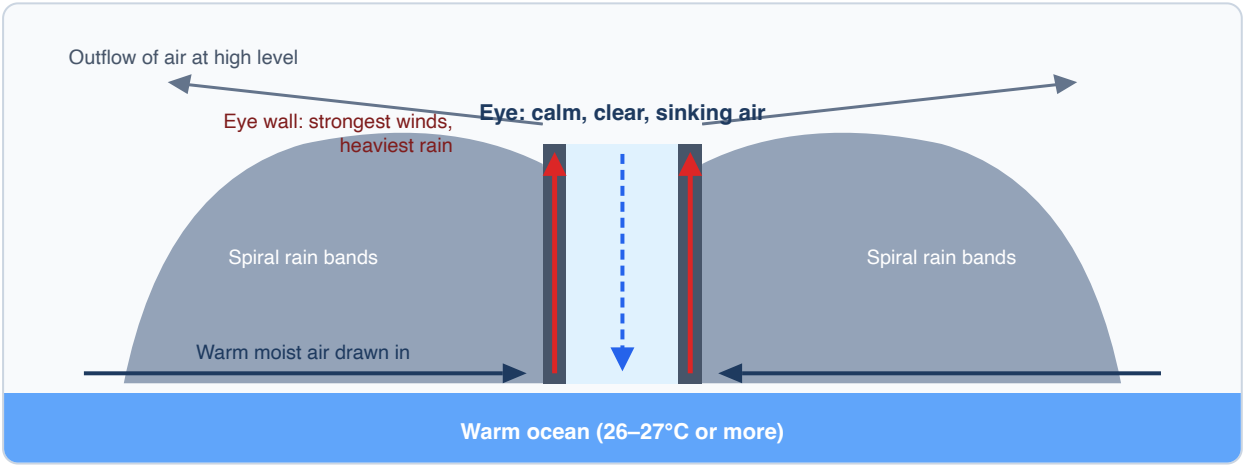
- Ocean surface temperature of **at least 26–27°C**, to a depth of about 50–60 m.
- Ocean **at least 5° from the Equator**, so the Coriolis effect can make the storm spin.
- **Low wind shear** — winds at different heights blowing at similar speeds, so the storm's structure is not torn apart.
- An area of **low pressure** with rising, unstable, moist air.

Sequence of formation

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1. Warm ocean water heats the air above it. The **warm, moist air rises rapidly**, creating an area of very low pressure at the surface.
2. As the air rises it **cools and condenses**, forming towering cumulonimbus clouds. Condensation releases **latent heat**, which warms the air further and makes it rise even faster — powering the storm.
3. Air rushes in at the surface to replace the rising air, creating strong winds. The **Coriolis effect** makes the inflowing air spiral (anticlockwise in the Northern Hemisphere).
4. Some cool, dry air sinks down the centre of the storm, creating the calm, cloud-free **eye**.
5. The storm is carried across the ocean by prevailing winds, growing as long as it has a supply of warm, moist air. It weakens when it reaches **land or cooler water** because the energy supply is cut off.

Structure of a tropical storm



Feature	Description
Eye	Centre of the storm, 10–50 km across. Calm, clear skies, lowest pressure, caused by sinking air.
Eye wall	Ring of towering cumulonimbus cloud around the eye — the most intense winds (often over 200 km/h) and heaviest rainfall.
Rain bands	Spiralling bands of cloud and rain extending hundreds of kilometres from the centre.

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Size	Typically 200–700 km across; can last for 1–3 weeks.

4. How Climate Change Might Affect Tropical Storms

Aspect	Likely change	Reason
Distribution	Storms may form further from the tropics (further north and south)	A larger area of ocean will reach 26–27°C
Frequency	Total number may stay the same or fall slightly	Scientists are uncertain — changes in wind shear may suppress some storms
Intensity	A higher proportion of storms reaching the strongest categories (4 and 5), with heavier rainfall	Warmer oceans supply more energy; warmer air holds more moisture
Storm surges	More damaging coastal flooding	Sea level rise means surges start from a higher base

Exam Tip: Be careful with certainty. The evidence for *more intense* storms is much stronger than the evidence for *more frequent* storms — say "likely" or "scientists predict" rather than "will".

5. Case Study: Typhoon Haiyan, the Philippines, November 2013

Typhoon Haiyan (known locally as Yolanda) — 8 November 2013

Background: The Philippines is a lower-middle-income country made up of over 7,000 islands in the western Pacific — one of the most typhoon-exposed countries in the world. Haiyan was one of the strongest tropical storms ever to make landfall, with sustained winds of about **315 km/h**. It hit the islands of Samar and Leyte, including the city of **Tacloban**.

Primary effects

- A **storm surge of up to 5–6 m** swept through Tacloban, flooding the city and its airport terminal.
- Over **6,300 people killed**; around 1,000 more missing; nearly 30,000 injured.
- About **1.1 million homes damaged or destroyed** — around 90% of Tacloban was destroyed.
- Around 4 million people displaced; power lines, roads and bridges destroyed.
- About 400 mm of rain caused widespread flooding.

Secondary effects

- About 14 million people affected; around 6 million workers lost their source of income.
- Flooding caused **landslides** and blocked roads, cutting off aid to remote areas.
- Shortages of food, water and shelter led to **outbreaks of disease** and some looting.
- Rice harvests and fishing fleets destroyed, causing food shortages; coconut plantations (a major source of income) flattened.
- An oil barge ran aground causing an oil spill, damaging mangroves.

Immediate responses

- The government evacuated around **800,000 people** before landfall, using warnings from the national weather agency (PAGASA).
- International aid and teams arrived — the UK sent military aircraft and HMS Daring; the USA sent the aircraft carrier USS George Washington to supply water and food.
- Field hospitals set up; emergency shelters, food, water and hygiene kits distributed by NGOs such as the Red Cross and Oxfam.
- A curfew was imposed two days after the storm to reduce looting.

Long-term responses

- A **"Build Back Better"** reconstruction plan — rebuilding roads, bridges and airports to a higher standard.
- "Cash for work" programmes paid local people to clear debris and rebuild, restoring incomes.
- Rice farmers and fishermen given help to restart — seeds, tools and replacement boats.
- A **"no-build zone"** proposed along vulnerable coasts, with communities relocated inland.

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- More **cyclone shelters** built; storm-surge warnings improved so people understood the risk (many did not know what a "storm surge" meant in 2013).
- Mangroves replanted along the coast to act as a natural barrier against future surges.

6. Reducing the Effects of Tropical Storms

Strategy	How it reduces the effects	Example
Monitoring	Satellites, aircraft and weather buoys track the storm's position, wind speed and pressure in real time	US National Hurricane Center tracks every Atlantic hurricane
Prediction	Computer models forecast the storm's path and strength several days ahead, allowing warnings and evacuation	PAGASA warnings allowed 800,000 to evacuate before Haiyan
Protection	Storm-surge barriers, sea walls, raised buildings on stilts, reinforced roofs, cyclone shelters	Bangladesh has built thousands of raised cyclone shelters, cutting death tolls dramatically since 1991
Planning	Evacuation routes, emergency supplies, education and drills; land-use zoning to keep homes away from the coast	Florida's signed hurricane evacuation routes; Philippines' no-build zones

Exam Tip: Planning and prediction save the most lives at the lowest cost — Bangladesh's 1970 Bhola cyclone killed around 300,000–500,000 people, whereas Cyclone Sidr (2007), of similar strength, killed about 3,400 thanks to early warnings and shelters. Use this contrast to evaluate strategies.

7. Weather Hazards in the UK

The UK's mid-latitude location, on the boundary between warm tropical and cold polar air masses and downwind of the Atlantic, means it experiences a range of weather hazards, although these are rarely as

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extreme as tropical storms.

Hazard	Cause	Example
Strong winds and storms	Deep Atlantic depressions	Great Storm of 1987; Storm Eunice (February 2022) — 196 km/h gust on the Isle of Wight
Heavy rain and flooding	A series of depressions; slow-moving fronts	Somerset Levels 2014; Storm Desmond, Cumbria (December 2015)
Heatwaves and drought	Persistent summer anticyclones (blocking highs)	2003, 2018 and 2022 heatwaves
Snow and cold spells	Cold polar or continental air; winter anticyclones	"Beast from the East" (February–March 2018)
Thunderstorms and hail	Intense summer heating causing rising air	Boscastle flash flood (2004)

8. Case Studies: Recent Extreme Weather in the UK

Case Study: Summer Heatwaves

2003 European Heatwave: France and southern UK experienced temperatures 6–7°C above average for over two weeks. UK recorded 38.5°C (Brogdale, Kent). Caused by a persistent anticyclone blocking Atlantic systems. Over 70,000 excess deaths across Europe; 2,000+ in UK. Crops failed; wildfires; power stations shut as river water temperatures rose too high for cooling.

2018 UK Heatwave: Prolonged high-pressure system from June to August. Temperatures reached 35.3°C (Faversham, Kent). UK's joint-hottest summer on record at the time. Drought conditions: rivers at record lows; hosepipe bans; moorland fires in Greater Manchester (Saddleworth Moor).

2022 UK Heatwave: On 19 July 2022, the UK recorded its highest ever temperature: **40.3°C** at Coningsby, Lincolnshire. Caused by a plume of hot air from North Africa channelled north by an anticyclone. Wildfires in London suburbs; rail lines buckled; Luton Airport tarmac melted; schools

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and businesses closed. Linked to climate change — Met Office stated such temperatures would have been virtually impossible without human-caused climate change.

Case Study: 2013–14 Somerset Levels Flooding

Winter 2013–14 was the UK's wettest in 248 years of records. A series of Atlantic depressions brought relentless storms. The Somerset Levels — a low-lying floodplain in SW England — were inundated from late December 2013 to March 2014.

- 65 sq km of farmland flooded; water depths up to 1 m; 17 million tonnes of water on the Levels
- Villages including Moorland and Muchelney were cut off for weeks by road
- Over 600 homes flooded and 16 farms evacuated; about 1,000 livestock moved to safety
- Somerset County Council initially estimated over £10 million of damage; a later economic impact assessment put the total cost to the county at up to £147.5 million
- **Environmental:** floodwater contaminated with sewage, oil and farm chemicals; debris deposited on fields; soils waterlogged for months
- **Causes:** a run of deep Atlantic depressions brought over 350 mm of rain in January–February 2014; the Levels are flat, low-lying (much of it at or below high-tide level) with clay soils, and high tides on the River Parrett stopped water draining away
- Controversy over the Environment Agency's decision to reduce dredging of rivers Tone and Parrett in prior years
- Long-term response: £100 million Somerset Levels Flood Action Plan; 8 km of rivers dredged; upgraded pumping stations; raised flood banks

Exam tip: When answering extreme weather questions, always include: specific dates, specific statistics (temperatures, cost, areas affected), social impacts AND economic impacts, and — for higher marks — link to the cause (e.g. blocking anticyclone, exceptional run of Atlantic depressions).

9. Is UK Weather Becoming More Extreme?

AQA asks you to consider the **evidence** that weather in the UK is becoming more extreme.

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Evidence	Detail
Temperature records	All ten of the UK's warmest years on record have occurred since 2002. The UK first passed 40°C in July 2022 (40.3°C, Coningsby).
Rainfall	Winter 2013–14 was the wettest on record for England and Wales. Storm Desmond (2015) set a UK 24-hour rainfall record of 341 mm at Honister Pass, Cumbria.
Flooding	Major floods have become more frequent: 2007, 2009, 2012, 2013–14, 2015–16, 2019–20 (Storms Ciara and Dennis).
Storms	The Met Office began naming storms in 2015 to raise public awareness of the risk. Several damaging named storms now occur each winter.
Drought	Hosepipe bans and drought declarations in 2018, 2022 and 2025.

Evaluating the evidence

- Records only go back to the mid-1800s, so a few decades of extreme events may be part of **natural variability**.
- Better measurement and media coverage mean extreme events are **reported more** than in the past.
- However, the Met Office's attribution studies show that climate change has made events such as the 2022 heatwave and the very wet winters **much more likely**. A warmer atmosphere holds about 7% more moisture per 1°C of warming, leading to heavier rainfall.

Exam Tip: For a 6-mark "To what extent is UK weather becoming more extreme?" question, give evidence for (records, frequency of floods and heatwaves) and against (short records, natural variability, better reporting), then reach a clear conclusion — e.g. "the trend in temperature extremes is clear, while the trend in storminess is less certain".

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